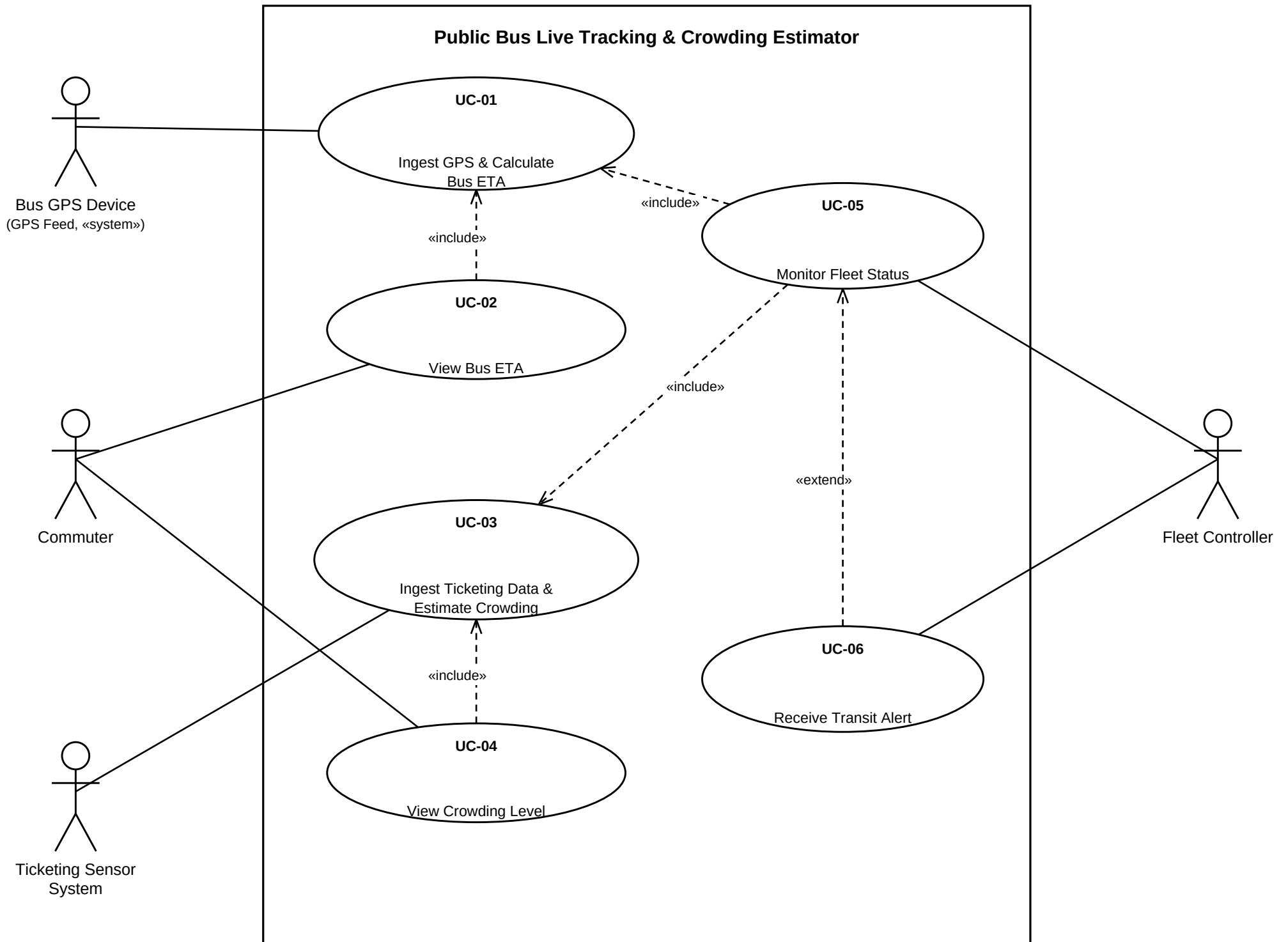


Public Bus Live Tracking & Crowding Estimator

Complete Requirements Table

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall ingest bus GPS coordinates every 5 seconds and recalculate arrival ETAs for all upcoming route stops.	High	Pass: ETA updates based on current traffic speed and distance. Fail: ETA increases while the bus approaches the stop.	Core capability the whole platform depends on; without live GPS and ETA recalculation no downstream feature (commuter view, fleet monitoring) is possible.
FR-002	Functional	The system shall allow a Commuter to select a bus route and stop and view the live location and ETA of the next 3 arriving buses, refreshed at least every 10 seconds.	High	Pass: selecting a stop displays at least 1 bus with a numeric ETA in minutes that refreshes within 10 s. Fail: screen shows no data or an ETA older than 10 s without a refresh indicator.	Directly serves the primary Commuter goal of the platform — knowing when a bus will arrive.
FR-003	Functional	The system shall compute a crowding level (Low / Medium / High) for each active bus at least once every 30 seconds using ticket validations from the onboard ticketing sensors against the bus's seating and standing capacity.	High	Pass: a bus with ticket count $\leq 60\%$ of capacity shows Low, 61–85% shows Medium, and $>85\%$ shows High, recalculated within 30 s of a new validation. Fail: crowding label does not change after a capacity-crossing validation event.	Crowding is one of the two headline outputs of the platform (with ETA) and directly supports the Commuter's decision of which bus/service to board.
FR-004	Functional	The system shall provide the Fleet Controller a live dashboard listing every active bus with its current location, ETA to its next stop, and crowding level, and shall raise an on-screen alert when a bus is delayed by more than 5 minutes versus schedule or reaches High crowding.	High	Pass: dashboard lists 100% of active buses with location, ETA and crowding columns populated, and an alert banner appears within 15 s of a bus crossing the 5-minute delay or High-crowding threshold. Fail: a threshold-breaching bus is not flagged within 15 s.	Enables the Fleet Controller's core responsibility of operational oversight and timely intervention across the fleet.
FR-005	Functional	The system shall detect when a bus's GPS feed or ticketing sensor feed has not updated for more than 30 seconds, mark that bus's data as “Stale/Unavailable” to Commuters and Fleet Controllers, and exclude it from ETA and crowding calculations until data resumes.	Medium	Pass: a bus with no GPS/sensor update for 30 s is labelled “Stale/Unavailable” within 5 s and disappears from ETA/crowding results. Fail: a bus with data older than 30 s continues to show a numeric ETA or crowding value.	Prevents the system from presenting misleading location, ETA, or crowding data to users when upstream telemetry fails, preserving trust in the platform.

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
NFR-001	Nonfunctional — Performance & Security	The tracking engine shall handle telemetry streams from up to 1,000 active transit vehicles simultaneously.	High	Pass: benchmarking tests confirm target latency and security standards under simulated peak load.	The platform is a city-wide service; it must scale to the full municipal fleet under peak-hour load without degrading ETA or crowding accuracy.
NFR-002	Nonfunctional — Data Freshness & Availability	The system shall ensure that any GPS or crowding data displayed to a Commuter or Fleet Controller is no more than 15 seconds old, and shall maintain at least 99.5% uptime for the live-tracking service measured monthly.	Medium	Pass: a monitoring log shows displayed-data age ≤ 15 s for $\geq 99\%$ of samples in a 24-hour test window, and monthly uptime $\geq 99.5\%$. Fail: displayed-data age exceeds 15 s for more than 1% of samples, or monthly uptime falls below 99.5%.	Complements NFR-001 (capacity/security) with a distinct, measurable freshness and reliability guarantee — stale or frequently-down tracking would defeat the purpose of a “live” tracking platform even if the system can technically handle 1,000 vehicles.



Use-Case Flow Specification

Public Bus Live Tracking & Crowding Estimator

Use Case Name	View Bus ETA
Use Case ID	UC-02
Primary Actor	Commuter
Supporting Actors	Bus GPS Device (GPS Feed) — via included use case UC-01: Ingest GPS & Calculate Bus ETA
Preconditions	1) The Commuter has the mobile app / web client open and has location or route-selection access. 2) At least one bus on the selected route is active and has transmitted a GPS update within the last 30 seconds (i.e., is not in “Stale/Unavailable” state, FR-005).
Postconditions	Success: the Commuter is shown the live location and ETA (in minutes) of the next arriving bus(es) for the chosen stop, current as of the last 10-second refresh cycle. Failure: the Commuter is shown a clear “ETA unavailable” state instead of a stale or incorrect ETA.

Main Success Scenario

1. The Commuter opens the app and selects a bus route and a stop (“Where is my bus?”).
2. The system retrieves the list of active buses currently assigned to that route (UC-01 has been continuously ingesting GPS every 5 s and recalculating ETAs in the background).
3. The system «includes» UC-01 (Ingest GPS & Calculate Bus ETA) to obtain the current position and freshest ETA for each candidate bus.
4. The system filters out any bus whose telemetry is older than 30 seconds (flagged Stale/Unavailable per FR-005) and ranks the remaining buses by ETA.
5. The system displays the next 3 arriving buses for the selected stop, each with a live map marker and an ETA in minutes.
6. The Commuter views the ETA and decides when to leave for the stop.
7. The system automatically refreshes the displayed ETA at least every 10 seconds while the screen remains open (FR-002).
8. The use case ends successfully when the Commuter closes the screen or navigates away.

Alternate Flow — 4a: GPS Data Becomes Stale / Unavailable

Trigger: at step 4, the selected route's only nearby bus has not sent a GPS update for more than 30 seconds.

- 4a1.** The system marks that bus's data as “Stale/Unavailable” and excludes it from the ETA list (per FR-005), instead of showing a frozen or incorrect ETA.
- 4a2.** The system displays the message “Live ETA temporarily unavailable for this bus — last known location shown” along with the timestamp of the last valid update.
- 4a3.** If another active bus on the same route has fresh telemetry, the system continues to display that bus's live ETA normally.
- 4a4.** If no bus on the route has fresh telemetry, the system shows “No live buses currently available on this route” and suggests the Commuter check the static timetable.
- 4a5.** The use case ends (either with a degraded but honest result, or resumes the Main Success Scenario automatically once GPS data resumes).